

## Los Angeles Regional Water Quality Control Board

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**AMENDMENT TO MONITORING AND REPORTING PROGRAM OF WASTE DISCHARGE REQUIREMENTS AND NATIONAL POLLUTANT DISCHARGE ELIMINATION SYSTEM (NPDES) PERMIT FOR LONG BEACH WATER RECLAMATION PLANT (WRP), JOINT OUTFALL SYSTEM (JOS) (NPDES PERMIT NO. CA0054119, ORDER NO. R4-2022-0032, CI-5662)**

Dear Katie Marjanovic:

NPDES Order No. R4-2022-0032 for the Long Beach WRP was adopted by the Los Angeles Regional Water Quality Control Board (Los Angeles Water Board) on February 10, 2022. Section 5 of the Monitoring and Reporting Program (MRP), Chronic Whole Effluent Toxicity Testing Requirements, requires JOS to select chronic toxicity methods based on the salinity of the receiving water, as follows:

1. If the effluent samples are collected from outfalls discharging to receiving waters with a salinity less than 1 part per thousand (ppt), JOS shall conduct chronic toxicity tests on effluent using species and test methods in *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms* (EPA/821/R02/013).
2. If the salinity of the receiving water is greater than or equal to 1 ppt, JOS shall conduct chronic toxicity tests on the receiving water using the species and test methods in *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to West Coast Marine and Estuarine Organisms* (EPA/600/R-95/136).

On June 2, 2025, the Los Angeles Water Board received a request from JOS to use freshwater species for chronic toxicity tests when the salinity at monitoring stations RSW-001 (approximately 970 feet upstream of the outfall) and RSW-002 (approximately 100 feet downstream of the outfall) are greater than 1 ppt. JOS included the following justification for this modification:

1. Salinity levels observed at stations RSW-001 and RSW-002 are not attributable to tidal influence and salinity averages are at or slightly below 1 ppt.

2. Using marine organisms would require adding salts to the test samples, which can adversely affect test organism sensitivity and compromise the reliability of test results.
3. It ensures consistency with Long Beach WRP final effluent toxicity monitoring, which is conducted using freshwater organisms, thereby facilitating a more reliable comparison of test results and improved identification of toxicity sources.

Los Angeles Water Board staff reviewed the information provided in the June 2, 2025, letter and concluded that the data set used to analyze the salinity and tidal height data for RSW-001 and RSW-002 was insufficient since the data was only collected between January 2024 and February 2025. On June 9, 2025, the Los Angeles Water Board staff requested that JOS include additional salinity and tidal height data covering a 5-year period. On October 1, 2025, JOS submitted additional salinity and tidal height data that was collected between August 2015 and August 2025. The Los Angeles Water Board staff reviewed the information provided by JOS and has determined the request to use freshwater species for chronic toxicity tests at the receiving water monitoring stations RSW-001 and RSW-002 is appropriate for the following reasons:

1. JOS conducted a salinity profile study in 2003, collecting salinity measurements at 15-minute intervals at RSW-001, RSW-002, and at the end of the concrete lining in the eastern low-flow channel in the San Gabriel River. The data from the salinity profile study indicate that the tide does not likely impact the salinity at station RSW-001, but the salinity at station RSW-002 may be impacted at tidal heights above 6.4 feet. Tidal height and salinity data collected between August 2015 and August 2025 indicate that the salinity at RSW-002 never exceeded 1 ppt when the tide exceeded the 6.4 feet. In fact, the highest tide recorded when the recorded salinity was greater than or equal to 1 ppt during this period was 4.43 feet. These results suggest that the salinity values reported for RSW-002 were likely not impacted by the tide during this period.
2. Considering the 119 salinity data points between August 2015 and August 2025, RSW-001 and RSW-002 generally exhibited freshwater conditions, with values averaging 0.24 ppt (ranging from 0.00 to 1.30) and 0.23 ppt (ranging from 0.00 to 3.00), respectively. In addition, salinity samples collected at RSW-001 and RSW-002 were equal to or slightly greater than 1 ppt 15.1% and 8.4% of the time, respectively. Based on requirements in section III.B.2. of the *State Policy for Water Quality Control: Toxicity Provisions* (Toxicity Provisions), marine test methods shall be used for receiving waters with salinity equal to or greater than 1,000 mg/L (1 ppt) at least 95 percent of the time; therefore, it is consistent with the Toxicity Provisions to use freshwater species and methods for these monitoring locations.
3. Conducting chronic toxicity tests on marine organisms in a low-salinity sample necessitates increasing the sample's salinity through the addition of artificial sea salt prior to testing. Based on documentation from the lab conducting the chronic toxicity tests on the sea urchin (the current most sensitive species), this addition of artificial sea salt may negatively affect the test organisms, alter the toxicity test outcomes, and compromise the reliability of the tests.

4. Utilizing the same test organism for both effluent and receiving water (upstream and downstream of the discharge point) facilitates a more effective comparison of toxicity results and identification of the source of downstream receiving water toxicity.
5. Section III.B.2. of the Toxicity Provisions stipulates that the permitting authority may require the use of freshwater test methods for dischargers who release freshwater effluent into marine waters or inland saline waters.

This decision aligns with the Toxicity Provisions and ensures the reliability and effectiveness of toxicity testing. Consistent with this decision, the first paragraph of section 5.3 and 5.4, and the last sentence of section 5.6.4 of the Monitoring and Reporting Program of Order No. R4-2022-0032, Chronic Freshwater Species and Test Methods, is hereby amended as follows (additions underlined and deletions with strikethrough text):

### 5.3. Chronic Freshwater Species and Test Methods

In 2003, the Discharger conducted a salinity profile study, where salinity measurements were collected at 15-minute intervals at RSW-001, RSW-002, and at the end of the concrete lining in the eastern low-flow channel in the San Gabriel River. The data from the salinity profile study indicate that the tide does not likely impact the salinity at station RSW-001, but the salinity at station RSW-002 may be impacted at tidal heights above 6.4 feet. Tidal height and salinity data collected between August 2015 and August 2025 indicate that the salinity at RSW-002 never exceeded 1 ppt when the tide exceeded the 6.4 feet. In fact, the highest tide recorded when the recorded salinity was greater than or equal to 1 ppt during this period was 4.43 feet. These results suggest that the salinity values reported for RSW-002 were likely not impacted by the tide during this period. In addition, based on the salinity samples collected between August 2015 and 2025, 8.4% of the samples at RSW-002 were equal to or greater than 1 ppt and 15% of the salinity samples collected at RSW-001 were equal to or greater than 1 ppt. Section III.B.3. of the Toxicity Provisions states, "...*marine test methods shall be used for receiving waters in which salinity is equal to or greater than 1,000 mg/L (1 ppt) at least 95 percent of the time.*" Since the salinity from samples collected at RSW-001 and RSW-002 do not exceed 1 ppt more than 95% of the time, it is consistent with the Toxicity Provisions to use freshwater methods for RSW-001 and RSW-002. If effluent samples are collected from outfalls discharging to receiving waters with salinity <1 ppt, the~~The~~ Permittee shall conduct the chronic toxicity tests on (1) effluent samples at the in-stream waste concentration for the discharge, (2) RSW-001, and (3) RSW-002 in accordance with species and test methods in *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms* (EPA/821/R02/013, 2002; Table IA, 40 CFR part 136). In no case shall these following species or test methods be substituted with other ~~test species or test methods~~ unless written authorization from the Executive Officer is received.

### 5.4. Chronic Marine and Estuarine Species and Test Methods

If the salinity of the receiving water sample collected at RSW-004, RSW-005, or RSW-006 is  $\geq 1$  ppt, the Discharger shall conduct the following chronic toxicity tests

on receiving water samples in accordance with the species and test methods in *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to West Coast Marine and Estuarine Organisms* (EPA/600/R-95/136, 1995). Artificial sea salts shall be used to increase sample salinity as needed. If the salinity of samples collected at RSW-004, RSW-005, or RSW-006 is <1 ppt, the Discharger shall discuss test method and species selection with the Los Angeles Water Board prior to test initiation. In no case shall these following species or test methods be substituted with other ~~test-species~~ or test methods unless written authorization from the Executive Officer is received.

- 5.6.4. If the receiving water chronic toxicity test does not meet all TAC specified in the referenced test method, *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving West Coast Marine and Estuarine Organisms* (EPA/600/R-95/136, 1995) (see Table E-5, below), then the Permittee must re-sample and re-test within 14 days. Any deviations from test specifications, specified in the referenced test method *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving West Coast Marine and Estuarine Organisms* (EPA/600/R-95/136, 1995), must be evaluated on a case-by-case basis to determine the validity of test results. The Discharger shall consider the degree of the deviation and the potential or observed impact of the deviation on the test results in consultation with Los Angeles Water Board staff before rejecting or accepting a test result as valid, and shall report the results of the validity determination with supporting evidence for that decision in their monthly report. ~~This applies when the salinity of the receiving water sample is  $\geq 1$  ppt.~~

If you have any questions, please contact Steven Webb at [Steven.Webb@waterboards.ca.gov](mailto:Steven.Webb@waterboards.ca.gov) or at (213) 576-6793, or Don Tsai at [Don.Tsai@waterboards.ca.gov](mailto:Don.Tsai@waterboards.ca.gov) or at (213) 576-6681.

Sincerely,

for Susana Arredondo  
Executive Officer  
Los Angeles Regional Water Quality Control Board